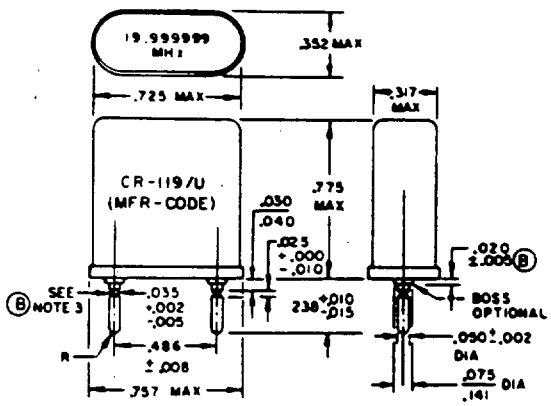
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-119/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.075	1.91
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
B 3. The pin undercut may be omitted.

B denotes change

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TABLE I. Requirements.

Requirement	Specified value or condition	
Holder	HC-6 U (MIL-H-10056 2)	
Dimensions, marking, and configuration	See figure	
Frequency range	0.8 to 20 MHz, inclusive	
Capacitance, shunt	7 pP, maximum	
Frequency tolerance	±50 PPM	
Equivalent resistance	See table II	
Antiresonance, load capacitance	30.0 ±0.5 pF	
Mode of oscillation	Fundamental	
Temperature range:		
Operating	-55° to +105°C. inclusive	
Test set, calibration values, and rated drive level	See table III	
Shock (specified pulse):	Below 2.0 MHz	2.0 MHz and above
Frequency change permitted	±10 PPM	±5 PPM
Equivalent resistance change permitted	±15 percent	±10 percent
Vibration: 1	Below 2.0 MHz	2.0 MHz and above
Frequency change permitted	±10 PPM	±5 PPM
Equivalent resistance change permitted	±15 percent	±10 percent
Temperature run:	Below 2.0 MHz	2.0 MHz and above
Frequency change permitted	±10 PPM	±5 PPM
Equivalent resistance change permitted	±15 percent	±10 percent
Aging:		
Frequency change permitted	5 PPM	

1 Method 201 of MIL-STD-202.

TABLE II. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.8 to 0.85	620
0.85 + to 0.9	600
0.9 + to 1	570
1 + to 1.12	540
1.12 + to 1.25	490
1.25 + to 1.37	450
1.37 + to 1.5	410
1.5 + to 1.62	370
1.62 + to 1.75	330
1.75 + to 1.87	300
1.87 + to 2	290
2 + to 2.12	270
2.12 + to 2.25	240
2.25 + to 2.6	190
2.6 + to 3	150
3 + to 3.4	110
3.4 + to 3.75	90
3.75 + to 4	75
4 + to 5	60
5 + to 7	35
7 + to 10	24
10 + to 15	22
15 + to 20	20

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TABLE III

EQUIVALENT RESISTANCE

Frequency Range Inclusive	Maximum Resistance
MHz	Ohms
1.85 to 1.999999	550
2.0 to 2.249999	500
2.25 to 2.999999	320
3.0 to 3.050000	175

NOTE: These crystal units were previously fabricated using Air Force Technical Order 12-1-20 and/or Collins Radio Company Specification 291-5969 thru 5970-00.

Custodians:
Air Force - 84

Review activities:
Air Force - 11, 17, 85

User activities:
Air Force - 19

Preparing activity:
Air Force - 84

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